

ABG INSURANCE PREDICTIVE ANALYTICS | DORMITORY RENTERS INSURANCE

Predictive Pricing Strategy

Transforming common premium structure into a risk-informed, coverage-specific pricing framework.

PRESENTED BY

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CONTEXT

CASE COMPETITION | October 2, 2026

EXECUTIVE SUMMARY

A data-driven roadmap to risk-aligned pricing and sustainable portfolio growth

KEY STRATEGIC OUTCOME: Coverage-specific tier pricing framework that enhances actuarial soundness, pricing transparency, and long-term profitability.

1

Accurate Risk Indication

Frequency-severity models across distinct coverages using approved rating factors.

2

Underwriting Alignment

Leverage 3 existing tiers while differentiating risk within each policy segment.

3

Technical Pricing

Convert expected loss to technical premium incorporating expenses and target profit.

4

Staged Deployment

Credibility-adjusted rollouts backed by strong governance and portfolio monitoring.

STRATEGIC CONTEXT

Transitioning from flat pricing to risk-informed rate structures

CORE ACTUARIAL PRINCIPLE

Claim frequency alone is insufficient — precise pricing requires modeling frequency and severity together.

⚠️ ABG's Current Challenge

- **Flat Premium Distortion:** Identical rates charged across diverse risk profiles.
- **Unpriced Risk Drivers:** Dormitory and student attributes generate varied claim loss costs.

🎯 Pricing Objective

Differentiate risk across four distinct coverages while maintaining operational simplicity and regulatory compliance.

- ✓ **Strategic Imperative:** Implement actuarially sound, transparent, and fair pricing anchored in predictive loss costs.

DATA FOUNDATION

Robust portfolio experience supporting predictive model calibration

40,071

RAW RECORDS

40,000

EXPOSURES

10,000

POLICYHOLDERS

1,748

CLAIM EVENTS

\$8.515M

TOTAL LOSSES

Permitted Rating Variables

- Prior claim history indicators
- Underwriting tier designations (Tier 1, 2, 3)
- Dormitory & property amenities
- Student attributes (academic standing, distance)

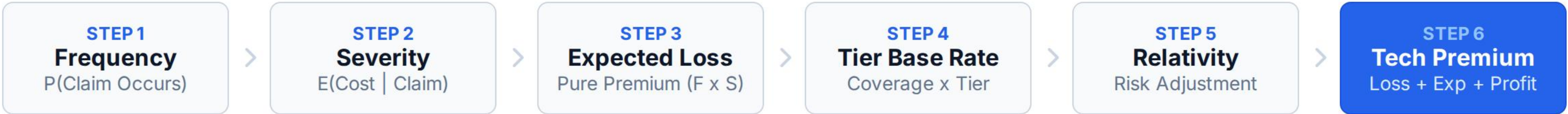
Data Pipeline

- Data scrubbing & exposure verification
- Categorical variable encoding
- Outlier treatment & distribution checks
- Dedicated holdout validation split

Data Expansion Path: Additional experience years | Fire protection attributes | Granular expense studies | Acquisition/retention tracking

METHODOLOGY

Actuarial frequency-severity modeling and rate calibration



Candidate Models Evaluated

Frequency Models: Logistic Regression, Calibrated Logistic, Random Forest, Gradient Boosting

Severity Models: Gamma GLM, Random Forest, Gradient Boosting

Model Selection Criteria

- Predictive performance metrics (AUC, PR-AUC, Brier score, RMSE, R²)
- Probability calibration & out-of-sample stability
- Actuarial interpretability & regulatory defense

VARIABLE STRATEGY

Actuarial inclusion rules ensuring ethical and regulatory alignment

✓ INCLUDED RATING FACTORS

Permitted risk predictors with actuarial support:

- **Underwriting Tier**
- **Property Features**
- **Student Academic Status**
- **Geographic Distance**

✗ EXCLUDED FROM PRICING

Protected attributes & non-rating variables:

- **Protected attributes** (gender retained solely for regulatory monitoring)
- **Policy identifiers**
- **Direct claim outcome indicators**

Continuous Variables Treatment: Modeled as continuous functions to maximize predictive power; evaluated for non-linear loss dynamics; operational rating bands designed for underwriting execution.

PORTFOLIO INSIGHTS

Distinct frequency and severity dynamics across coverage lines

Coverage Line	Claim Frequency	Average Severity	Observed Pure Premium
Personal Property	9.22%	\$1,080	\$99.55
Additional Living Expense (ALE)	5.12%	\$5,686	\$291.13
Guest Medical	2.29%	\$16,902	\$387.06
Liability	0.85%	\$8,675	\$73.74

KEY ACTUARIAL INSIGHT: Personal Property exhibits the highest claim frequency, while Guest Medical generates the largest average severity. Stand-alone coverage pricing is necessary to capture these distinct risk drivers.

MODEL PERFORMANCE

Holdout backtesting confirms strong calibration except for low-volume Liability

Coverage	Pred/Actual Loss	Freq. AUC	Sev. R ²
Personal Property	1.072	0.554	0.393
Additional Living Expense	0.987	0.591	0.447
Guest Medical	1.131	0.565	0.236
Liability	0.682	0.685	-0.028

Validation Takeaways

- **ALE** achieves near-perfect aggregate calibration (0.987).
- **Personal Property & Guest Medical** fall within acceptable $\pm 15\%$ tolerance.
- **Liability** underpredicts losses (68.2%) due to small claim counts, requiring credibility adjustments.

TIER DYNAMICS

Observed loss experience suggests increasing loss cost across the supplied tiers

Underwriting Tier	Claim Frequency	Average Severity	Pure Premium
Tier 1	4.62%	\$4,136	\$191
Tier 2	4.45%	\$4,970	\$221
Tier 3	3.97%	\$5,973	\$237

KEY ACTUARIAL INSIGHT: Observed pure premium increases from approximately \$191 in Tier 1 to \$237 in Tier 3, despite declining claim frequency. The difference is primarily severity-driven, reinforcing the need to evaluate both claim likelihood and claim magnitude.

STRUCTURE

PRICING ARCHITECTURE

Converting expected loss estimates into final technical premiums

BASE RATE

Tier Base Premium

coverage x tier

X

RELATIVITY

Policy Factor

policy EL / tier base

=

EXPECTED LOSS

Policy Loss Cost

base x relativity

+

LOADINGS

Expenses & Profit

20% Exp + 10% Profit

=

FINAL RATE

Technical Premium

Per Coverage

Technical Rating Formula

Technical Premium = (Expected Loss + Fixed Expense) / (1 - Var Expense - Profit)

Individual coverage premiums are aggregated to form total policy premium.

Illustrative Calculation (ALE Tier 2)

- Tier Base Pure Premium = \$277.74
- Policy Relativity Factor = 1.20
- Expected Policy Loss = \$277.74 × 1.20 = \$333.29
- Technical Premium = \$333.29 / (1 - 0.30) = \$476.13

REFINEMENTS

ACTUARIAL ADJUSTMENTS

Credibility weighting applied to stabilize low-volume Liability claims

67	0.682	0.418	0.422	0.989
DEV CLAIMS	HOLDOUT PRED/ACTUAL	FREQ. CREDIBILITY (Z)	SEV. CREDIBILITY (Z)	SEVERITY CV

FREQUENCY CREDIBILITY FORMULA

Pricing Frequency = 0.418 x (Model Frequency) + 0.582 x (Portfolio Benchmark Frequency)

SEVERITY CREDIBILITY FORMULA

Pricing Severity = 0.422 x (Model Severity) + 0.578 x (Portfolio Benchmark Severity)

Actuarial Outcome: Credibility blending prevents rate volatility for low-frequency Liability coverage while preserving policy-level risk differentiation.

OPERATIONAL IMPLEMENTATION

Phased rollouts for new business and renewal policyholders

New Business Strategy

- Pure model-led rating due to lack of prior customer history.
- Acquisition cost provisions tailored to new customer channels.
- Tier designation and rating factors establish baseline pricing.

Renewal Business Strategy

Individual claim history incorporated as policy credibility develops.

$$\text{Renewal Rate} = Z \times (\text{Observed Loss}) + (1-Z) \times (\text{Model Indication})$$

Expense loadings adjusted for lower renewal acquisition costs.

1. Expense Allocation

Reflect true acquisition and claims overhead in expense loadings.

2. Tier Offsets

Ensure manual credits/debits align with modeled relativities.

3. Retention Monitoring

Track quote conversion and policyholder retention post-launch.

STRATEGIC OUTLOOK & ACTION PLAN

Illustrative 3-Year Portfolio Outlook and strategic recommendations for leadership

Illustrative 3-Year Portfolio Outlook

Metric	Year 1	Year 2	Year 3
Policyholders	2,101	2,206	2,316
Earned Premium	\$2.590M	\$2.801M	\$3.029M
Incurred Losses	\$1.831M	\$1.999M	\$2.183M
Loss Ratio	70.7%	71.4%	72.1%

Assumptions: 5% policy growth | 4% annual loss trend | 3% rate change adjustment

CEO Recommendations

1. Adopt coverage-specific predictive pricing through a controlled phased rollout.
2. Enforce Liability credibility weighting to stabilize initial rate changes.
3. Execute phased rollout with monthly retention and rate-adequacy monitoring.
4. Recalibrate frequency and severity models annually with emerging data.

APPENDIX A: FREQUENCY MODELS

Validation metrics across candidate frequency model specifications

Coverage	Selected Model	AUC	PR-AUC	Brier	Observed Freq.	Predicted Freq.
ALE	Logistic	0.599	0.080	0.0521	5.55%	5.13%
Guest Medical	Random Forest	0.619	0.079	0.0233	2.40%	2.26%
Liability	Calibrated Logistic	0.599	0.011	0.0084	0.85%	0.85%
Personal Property	Logistic	0.584	0.144	0.0819	9.15%	9.43%

Methodological Note: Model selection emphasized probability calibration and PR-AUC over simple accuracy due to low event frequencies in personal lines renters data.

APPENDIX B: SEVERITY MODELS

Diagnostics for Gamma GLM severity models selected across coverages

Coverage Line	Claims Count	MAE	RMSE	R ² Score	Actual Severity	Predicted Severity
ALE	416	\$2,721	\$4,229	0.470	\$5,912	\$5,323
Guest Medical	183	\$11,928	\$18,712	0.345	\$19,224	\$17,045
Liability	67	\$8,907	\$13,172	-0.266	\$13,210	\$7,309
Personal Property	743	\$540	\$814	0.361	\$1,098	\$1,061

Diagnostic Note: Liability severity modeling is limited by small sample size (67 claims). This volatility is addressed via credibility weighting rather than standalone unadjusted indications.

APPENDIX C: TIER INDICATIONS

Coverage-specific pure premium and technical premium rates by tier

PURE PREMIUM INDICATIONS

Coverage	Tier 1	Tier 2	Tier 3
ALE	\$248.55	\$277.74	\$306.86
Guest Medical	\$330.60	\$420.54	\$396.45
Liability	\$58.00	\$60.13	\$71.32
Personal Property	\$97.23	\$105.69	\$91.09

TECHNICAL PREMIUM INDICATIONS*

Coverage	Tier 1	Tier 2	Tier 3
ALE	\$355.07	\$396.77	\$438.37
Guest Medical	\$472.29	\$600.78	\$566.35
Liability	\$82.86	\$85.91	\$101.88
Personal Property	\$138.90	\$150.99	\$130.13

*Reflects 20% Variable expense and 10% target profit loading; excludes fixed expense fee components.

ABG INSURANCE PREDICTIVE PRICING STRATEGY

Thank You For Your Time & Review

ABG Insurance Predictive Analytics Division - Dormitory Renters Insurance



CASE COMPETITION | OCTOBER 2, 2026

METHODOLOGY & ACKNOWLEDGMENTS

Analytical stack, software, and supplemental communication resources



Python

PRIMARY ANALYTICAL ENGINE

Data preparation, statistical modeling, validation, credibility analysis, and scenario testing.



Microsoft Excel

AUDIT & VALIDATION

Results review, actuarial cross-checking, and numerical validation.



PowerPoint

PRESENTATION DEVELOPMENT

Presentation layout, visual structuring, and stakeholder communication collateral.



ChatGPT

SUPPLEMENTARY RESOURCE

Presentation refinement, technical concept clarification, and communication support.

GOVERNANCE & OWNERSHIP STATEMENT: All analytical decisions, model selection, interpretation of results, and recommendations were made and reviewed by the team.